

ESPRIT · 2026

SOFTWARE ENGINEERING

PEAXIS

A Modular AI-Powered Hiring Operating System

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Next.js

NestJS

FastAPI

PostgreSQL

pgvector

Gemini

Stripe

Docker

1 INTRODUCTION & CONTEXT

Presentation Plan

A roadmap for a 20–25 minute engineering defense.

01 Introduction & Context

Internship context at Prospector, role background, and PFE goals.

02 Problem Analysis

Recruitment bottlenecks, legacy ATS constraints, and market comparisons.

03 Methodology & Requirements

Software engineering lifecycle, functional goals, and SLA constraints.

04 Proposed Solution

PEAXIS ecosystem overview: Core governance, Hire ATS, and Jobs portal.

05 Architecture & Engineering

Technology stack, logical and physical architecture, AI pipeline, security, data, and deployment decisions.

06 Results & Close

Engineering achievements, technical roadmap, defense conclusion, and jury discussion.

1 INTRODUCTION & CONTEXT

Why PEAXIS?

Hiring teams need one operating layer, not another isolated tool.

01 Hiring Fragmentation

Spreadsheets, calendars, job boards, and CV folders run separately.

02 Legacy ATS Failure

Keyword filters miss non-standard CVs and weakly explain decisions.

03 Candidate Friction

Applicants get little feedback, status visibility, or skill-gap guidance.

04 The Intelligence Gap

AI should be measurable infrastructure, not a surface feature.

- **Core thesis:** PEAXIS unifies business operations, recruiter execution, candidate experience, and explainable AI in one **modular hiring OS**.

1 INTRODUCTION & CONTEXT

Prospecter — Internship Host

AI-powered B2B sales prospecting platform where I served as **Technical Lead & Software Engineer**.

MISSION

Autonomous outbound lead generation for scalable B2B revenue teams.

AI / LLM

B2B SAAS

SALES AUTOMATION

OUTBOUND GTM



AI SDR Engine

Qualifies leads and orchestrates outreach sequences at scale.



Lead Intelligence

Prioritises accounts using firmographics and intent signals.



Outbound Automation

Multi-channel outreach with LLM-personalised messaging.



Pipeline Analytics

Conversion tracking, message tests, and CRM sync.



Lead Engineer Synergy: Production exposure to **LLM pipelines**, **tenant isolation**, and **async queues** directly informed PEAXIS architecture.

1 INTRODUCTION & CONTEXT

Building PEAXIS at Production Scale

Independent hiring operating system built as the internship project.

Scope & Integrity: Independent from Prospector's codebase, built with academic rigor and SaaS production standards.



Full-Stack Architecture

- NestJS API: 14 modules
- Next.js: Core, Hire, Jobs, Landing
- Tenant-scoped data model



AI Integration

- FastAPI AI service
- CV parsing + match scoring
- Copilot, JD generation, bias check



Platform Engineering

- Docker deployment
- pgvector semantic search
- Stripe billing + webhooks

TECHNICAL LEAD

SOFTWARE ENGINEER

AI INTEGRATION

PLATFORM ARCHITECT

SOLO CONTRIBUTOR

2 PROBLEM ANALYSIS

Recruitment is **broken** at every layer

Slow workflows, manual screening, fragmented tools.

42 days

Avg time-to-hire
SHRM 2024

75%

Recruiters overloaded
LinkedIn Talent Report

60%

Poor UX ghosting
Indeed Survey 2023



Screening bottleneck

Manual CV review consumes recruiter time and slows every pipeline.



Bias & inconsistency

Criteria vary by reviewer, increasing bias risk and weak auditability.



Candidate friction

Long forms, no feedback, and unclear status push talent away.

Global Cost: Bad hires average **\$14.9K**; strong candidates move in days.

Regional Gap: MENA SMEs still rely on spreadsheets because legacy ATS tools are costly and poorly localized.

2 PROBLEM ANALYSIS

Benchmarking PEAXIS

PEAXIS balances modular SME infrastructure with enterprise-level explainable intelligence.

CAPABILITY	GREENHOUSE	LEVER	WORKABLE	ASHBY	HIREVUE	EIGHTFOLD AI	PEAXIS
Core ATS	✓	✓	✓	✓	✗	partial	✓
Candidate Portal	basic	basic	basic	✗	✓	✗	✓
Explainable AI	✗	✗	partial	✗	partial	✓	✓
Integrated Ecosystem	✗	✗	partial	partial	✗	✗	✓
Tenant Isolation	✓	✓	✓	✓	✓	✓	✓

Strategic Position: PEAXIS combines ATS workflows, candidate applications, and **explainable AI scoring** in an SME-focused product.

2 PROBLEM ANALYSIS

The market gap: **integrated, explainable AI**



AI black boxes

Recommendations need traceability: requirements, cited evidence, and recruiter review.



Fragmented tooling

Specs, pipelines, calendars, and notes live in separate tools.



No candidate intelligence

Tools track applications but rarely guide decisions or interviews.

\$1.2B

HR Tech VC funding in 2024

23%

CAGR of AI Recruitment market

78%

HR leaders plan AI adoption in 2025



PEAXIS fills this exact gap

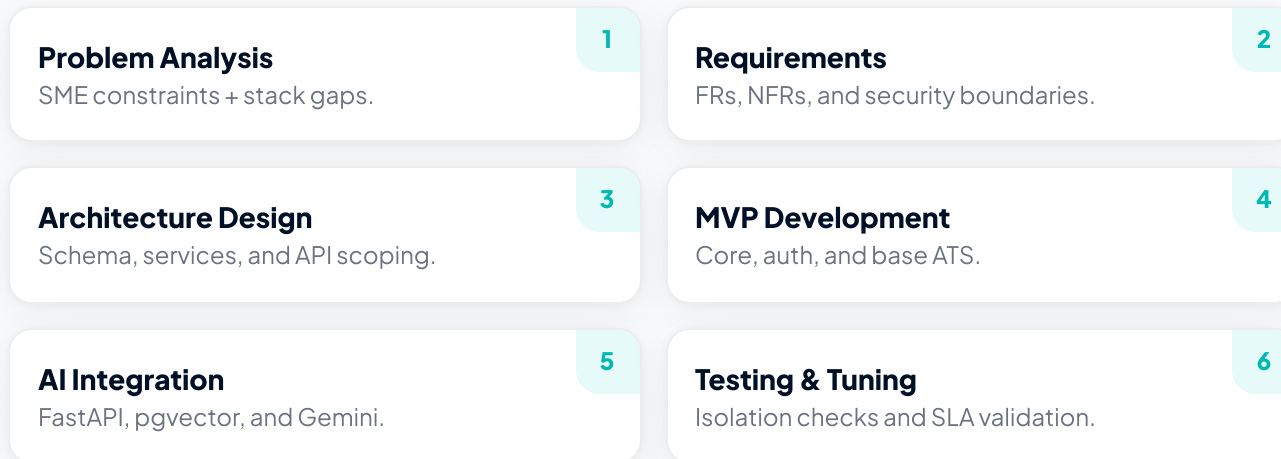
Explainable scores, integrated workflows, and a candidate-first experience in one OS layer.

3 METHODOLOGY & REQUIREMENTS

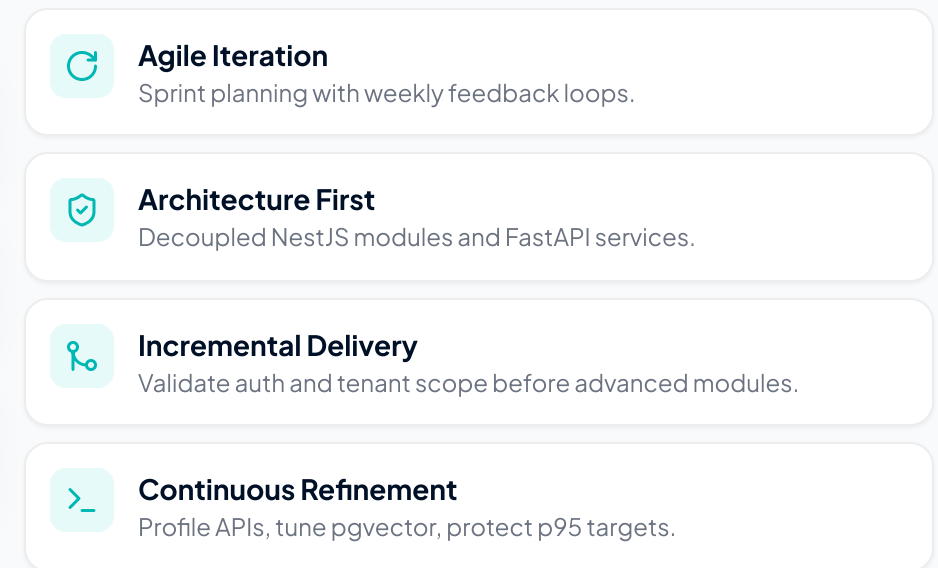
Engineering Methodology

Translating academic requirements into a production-grade modular product.

PROJECT ENGINEERING PIPELINE



EXECUTION STRATEGIES



3 METHODOLOGY & REQUIREMENTS

Core platform requirements

FR-01 to FR-05: the foundation for a production-grade recruitment SaaS.



FR-01

Authentication & Security

- Argon2 password hashing
- Google + LinkedIn OAuth2
- 15m access + 7d refresh JWT



FR-02

Multi-Tenant Architecture

- Business-scoped queries
- X-Business-ID enforcement
- Unique business slug routing



FR-03

Roles & Permissions

- Plan-based feature gating
- Free / Starter / Pro / Enterprise
- JwtAuthGuard + PlanGuard



FR-04

Job Management

- CRUD + rich filters
- Semantic search via pgvector
- Automatic embeddings



FR-05

Application Pipeline

- Applied to Hired/Rejected
- Stage transition history
- AI score at submission

3 METHODOLOGY & REQUIREMENTS

AI & analytics requirements

FR-06 to FR-10: the intelligence layer above traditional ATS workflows.



FR-06

AI Match Scoring

- Skill-overlap engine
- 0–100 score + explanation
- Strength/weakness breakdown



FR-07

CV Parsing

- PDF / DOCX / TXT upload
- Configured provider extraction
- Durable candidate-profile parsing



FR-08

Candidate Onboarding

- Apply with CV
- Profile completeness score
- Candidate embeddings



FR-09

Interview Scheduling

- Interview CRUD + type
- Application-linked schedule
- Interviewer assignment



FR-10

Analytics Dashboard

- Pipeline funnel metrics
- Score distribution charts
- Time-to-hire tracking

3 METHODOLOGY & REQUIREMENTS

Production-grade **quality attributes**



Horizontal scale

NFR-01 **Scalability**

Stateless API pods; Redis queues absorb spikes.



OWASP aligned

NFR-02 **Security**

Argon2, JWT rotation, OAuth2, tenant scoping, OWASP checks.



p95 < 200ms

NFR-03 **Performance**

Redis caching + BullMQ keep heavy work off requests.



99.5% SLA target

NFR-04 **Availability**

SDK retries, job retries, and Docker health checks.



Full audit trail

NFR-05 **Explainability**

Scores include match, label, skills, gaps, and explanation.



Microservices

NFR-06 **Modularity**

NestJS modules, FastAPI service, decoupled frontends.

4 PROPOSED SOLUTION

PEAXIS — Modular AI Hiring Operating System

Three integrated modules. One ecosystem, one data layer, every stakeholder served.



Operational Backbone

PEAXIS Core

Auth, billing, tenant governance, roles, and accounts.



Recruiter Execution

PEAXIS Hire

ATS pipeline, scoring, summaries, interviews, and analytics.



Talent Acquisition

PEAXIS Jobs

Semantic search, CV parsing, match preview, and easy apply.

BUSINESS ADMIN

PEAXIS Core

Governance, billing, teams, settings

RECRUITER / HR TEAM

PEAXIS Hire

Pipelines, scoring, analytics, collaboration

CANDIDATE / TALENT

PEAXIS Jobs

Search, match preview, easy apply

• All modules share a single Prisma → PostgreSQL data layer

NEXT.JS 16

NESTJS

FASTAPI

POSTGRESQL + PGVECTOR

REDIS

BULLMQ

GEMINI

STRIPE

4 PROPOSED SOLUTION

PEAXIS Core — Business Operations

Business infrastructure for authentication, tenancy, billing, and team operations.



Authentication & Security

JWT RS256 +
Google/LinkedIn OAuth



Multi-Tenant Isolation

businessId scoping on
org data



Role-Based Access

Owner, Admin, Recruiter,
Viewer



Billing & Subscriptions

Stripe subscriptions +
webhooks



Team Management

Seats, settings,
preferences



Account Governance

Users, logs, security
policy

Strategic Value: One business control center for teams, plans, access, and governance.

JWT RS256

MULTI-TENANT

STRIPE BILLING

ROLE-BASED ACCESS

4 PROPOSED SOLUTION

PEAXIS Hire — Recruiter Execution

Core ATS platform for AI-assisted recruiter execution.

Workspace & Pipeline

-  **Job Workspace**
Pipeline, analytics, and job configuration.
-  **Pipeline Kanban**
Applicant tracking with inline AI scoring.
-  **Candidate Drawer**
Full profile review without leaving context.
-  **Offer Management**
Offer drafts, scheduling, and sign-offs.

Recruiter Intelligence

-  **AI Match Scoring**
Required skills vs candidate skills.
-  **Recruiter AI**
Evidence review, summaries, and interview-question drafts.
-  **Decision Analytics**
Funnel metrics and SLA status.
-  **Skill Gap Tool**
Strengths and weaknesses by skill.

Collaboration & Flow

-  **Interview Scheduler**
Calendar slots and interview types.
-  **Team Approvals**
Role-based feedback and approvals.
-  **Activity & Notes**
Notes, chat, and decision history.
-  **Candidate Profiles**
Private resumes and portfolio data.

Strategic Value: One audit-ready recruiter interface for tracking, collaboration, and evaluation.

4 PROPOSED SOLUTION

PEAXIS Jobs — Talent Acquisition

Candidate portal for discovery, matching, and low-friction applications.

Job Discovery & Search

-  **Semantic Job Search**
Embedding search beyond keywords.
-  **Advanced Filters**
Skills, salary, location, remote.
-  **Job Lite Format**
Fast cards for mobile browsing.
-  **Personalised Feed**
Recommendations by candidate fit.

Application UX

-  **AI CV Parsing**
CV upload fills profile fields.
-  **Easy Apply Flow**
Parsed CV + profile in one step.
-  **Pipeline Tracking**
Status visibility by stage.
-  **Profile Manager**
Preferences, portfolio, skills.

AI Matching & Insights

-  **Semantic Discovery**
Hybrid search helps candidates discover relevant jobs.
-  **Skill Gap Tool**
Strengths + missing skills.
-  **Job Bookmarks**
Save roles for later.
-  **Job Alerts**
Notifications for matching roles.

Strategic Value: Replaces opaque applications with clear matching insights and faster apply flows.

4 PROPOSED SOLUTION

PEAXIS AI Brain — Intelligence Layer

Stateless inference service with API-owned evidence, authorization, and hiring decisions.

Recruiter AI

- Evidence-based assessment
- Cited requirement evaluations
- Recruiter content artifacts
- Audited review and overrides

Candidate AI

- Durable CV parsing
- Application tracking
- Public job discovery

Employer AI

- Requirement definition
- JD generation drafts
- Usage and credit controls



Evidence-first: Requirements, cited evidence, evaluations, and recruiter review remain auditable.



Durable work: BullMQ and the dedicated worker keep parsing and assessment outside API requests.

FASTAPI

GEMINI / AZURE OPENAI

PGVECTOR

REDIS CACHE

5 ARCHITECTURE & TECHNOLOGIES

Tech Stack — Four Layers

Production-grade architecture. Modular, scalable, and designed for real-world hiring at scale.

Frontend

Core, Hire, Jobs, and Landing apps with shared React/TypeScript patterns

Next.js 16

React

Tailwind

TypeScript

API

NestJS + Prisma: 14 modules, 60+ endpoints, one data authority

NestJS

Prisma

JWT

TypeScript

AI Service

Stateless inference for parsing, classification, embeddings, and structured generation

FastAPI

Gemini / Azure

Python

Redis

Database

Authoritative data + pgvector retrieval, Redis, and one durable BullMQ queue

PostgreSQL

pgvector

Redis

BullMQ

MULTI-TENANT

Every query scoped to businessId

ASYNC PROCESSING

BullMQ + background workers

RETRIEVAL

pgvector supports job discovery and cited evidence

5 ARCHITECTURE & TECHNOLOGIES

Logical Architecture

Clear ownership: applications present, the API governs, the worker executes, and FastAPI infers.



Next.js applications

Jobs, Hire, Core, and Admin present workflows; they do not own authorization or AI decisions.



NestJS API

JWT, tenant access, entitlements, orchestration, and every authoritative write.



Dedicated api-worker

Consumes durable AiWorkItem records from the single BullMQ ai-tasks queue.



FastAPI inference boundary

Validates requests and calls the configured Gemini or Azure OpenAI provider; no database ownership.



PostgreSQL + pgvector

Authoritative operational data, evidence assessments, audit records, and retrieval vectors.

5 ARCHITECTURE & TECHNOLOGIES

Physical Architecture

Docker Compose separates request handling, async execution, inference, and stateful services.



Web applications

Core, Hire, Jobs, Admin



API

NestJS HTTP runtime; AI workers disabled



api-worker

Dedicated BullMQ ai-tasks consumer



AI service

FastAPI, internal-only inference endpoint



Data services

PostgreSQL + pgvector and Redis



Trust boundary: only the API and worker call FastAPI; FastAPI has no PostgreSQL credentials.

5 ENGINEERING DEEP DIVE

Architecture Decisions

The stack was selected for delivery risk, maintainability, and production behavior.

Modular monolith

vs microservices

Domain stays cohesive; less distributed complexity.

Separate AI service

FastAPI

Python ML stack isolated from core API load.

Next.js apps

3 frontends

Server rendering plus rich client interactions.

PostgreSQL

Prisma

Relational workflows with strong consistency.

BullMQ

async

Heavy AI work moves outside request lifecycle.

pgvector

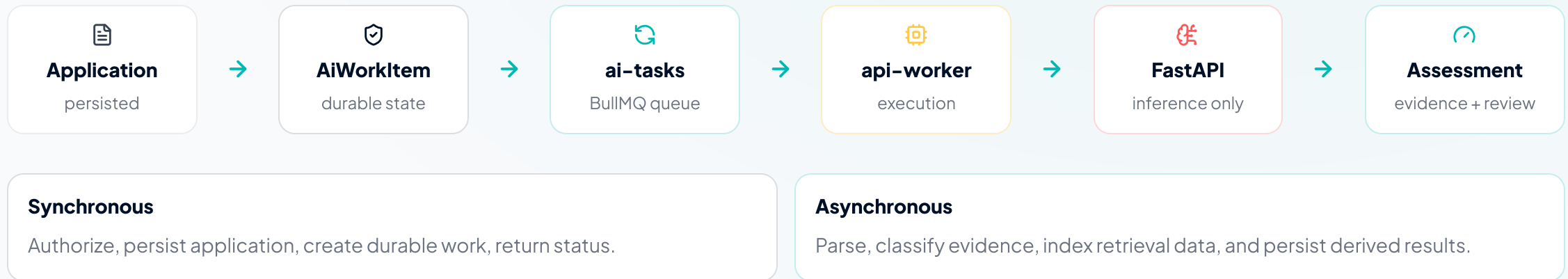
retrieval

Retrieval for job discovery and cited evidence.

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End-to-End AI Pipeline

Business data is committed first; durable AI work runs in a dedicated worker.



5 ENGINEERING DEEP DIVE

Evidence Assessment

Inference is bounded by explicit requirements, cited evidence, and human review.

Requirements

Employers define job-relevant criteria and verification policies.

Inference

Configured provider extracts or classifies structured evidence.

Persistence

Stores evaluations, cited evidence, and audited reviewer actions.

Requirement



Evidence



Evaluation



Review



Recruiter action

5 ENGINEERING DEEP DIVE

Assessment Controls

The system records what supports each requirement and what needs verification.

Requirement

input

Stable, job-relevant criterion defined by the employer.

Evidence

traceable

Claim or source chunk cited for the evaluation.

Verification

human

Recruiter confirmation or external check when required.

Assessment = requirement + cited evidence + evaluation + reviewer action

Evidence claim

Needs verification

Recruiter override

Audit trail

5 ENGINEERING DEEP DIVE

Explainable AI

Recruiters see the evidence behind the recommendation, not only a number.

EVIDENCE ASSESSMENT

Frontend Engineer requirements

READY
REVIEW

Supported

Evidence cites React and TypeScript project experience.

Verify

Confirm GraphQL and work-authorization requirements.

Requirement

What the role requires.

Evidence

Why the evaluation is supported.

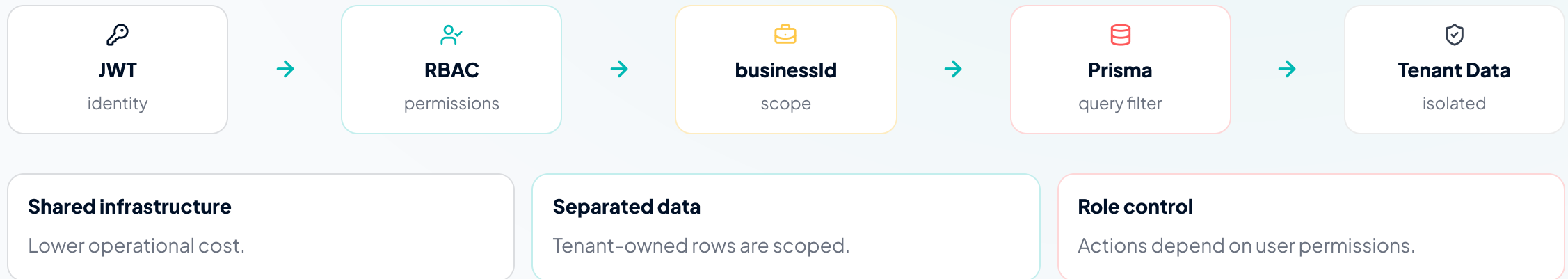
Review

Recruiter confirms, waives, or overrides.

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Multi-Tenant Architecture

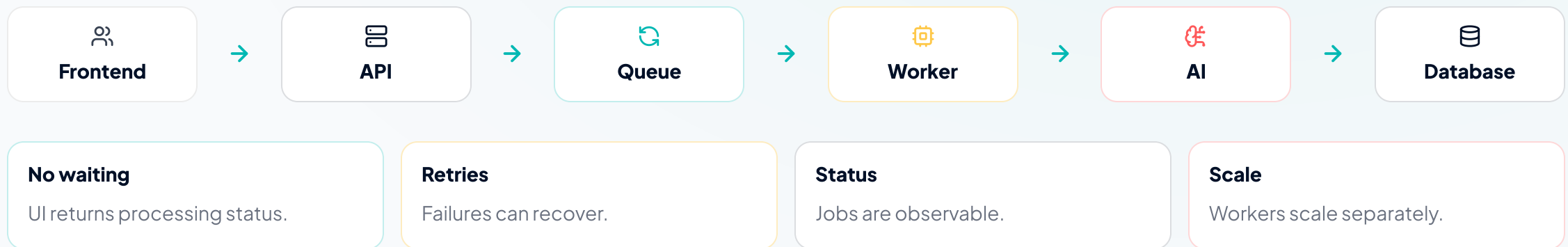
One shared platform, separated business data, and request-level tenant isolation.



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Background Processing

BullMQ keeps AI work reliable without blocking recruiters or candidates.



5 ENGINEERING DEEP DIVE

Caching Strategy

Redis reduces repeated AI cost and improves response time on expensive paths.

Versioned keys

Model and schema versions avoid incompatible cache reuse.

Bounded results

Safe repeated inference can reuse cached output.

Not a source of truth

PostgreSQL retains the authoritative platform state.



Cost

Fewer duplicate model calls.



Latency

Hot paths skip AI work.



Consistency

Stable inputs return stable outputs.

5 ENGINEERING DEEP DIVE

Security Architecture

Security is layered across identity, authorization, tenant scope, and service boundaries.



JWT

Access token authorization



Refresh tokens

Session continuity



OAuth

External login providers



Argon2

Password hashing



RBAC

Role-based actions



Rate limit

Abuse protection



Secrets

Service-to-service trust



OWASP

Validation and hardening

5 ENGINEERING DEEP DIVE

Database Design

PostgreSQL models the hiring workflow, while pgvector adds semantic search.

Business



Jobs



Applications



Candidates



AI Results



Indexes

businessId, status, relations, search keys.



Vectors

Retrieval support for public job search and cited evidence.



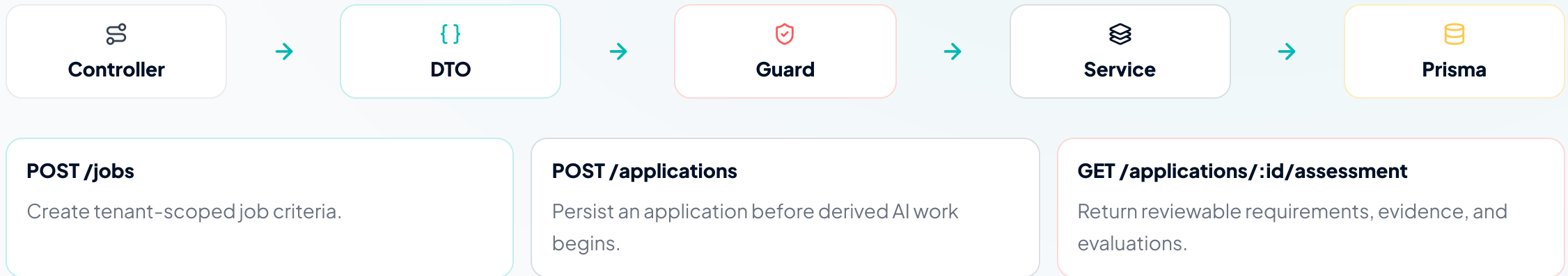
Audit

Stored AI evidence and workflow history.

5 ENGINEERING DEEP DIVE

REST API Design

NestJS modules keep HTTP contracts, validation, policy, business logic, and persistence separated.



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AI Models & Routing

One configured provider is selected per deployment; it never silently falls back.

Feature	Model route	Reason
CV parsing	Gemini or Azure OpenAI	Structured extraction
Evidence classification	Configured provider	Requirement-focused inference
Generated content	Configured provider	Recruiter-reviewed artifact
Embeddings	Configured deployment	Retrieval infrastructure
Assessment	NestJS + PostgreSQL	Evidence and human review

Route by task

Cache repeated inputs

Control latency

Control cost

5 ENGINEERING DEEP DIVE

Performance Optimizations

The platform improves perceived speed by removing repeated work and isolating slow operations.



Server Components

Less client JavaScript



Redis

Hot cache for expensive results



BullMQ

AI outside request path



Prisma indexes

Predictable DB queries



Parallel fetching

Faster dashboards



Workers

Independent throughput

Goal: fast UI, reliable AI, scalable workers.

Technical Challenges & Solutions

The project required engineering choices for cost, delay, isolation, search, and deployment.

Challenge	Solution	Outcome
AI delay	BullMQ workers	Responsive UX
LLM cost	Redis cache	Lower repeat spend
Tenant isolation	businessId + RBAC	Secure SaaS
Keyword search	pgvector	Semantic recall
Explainability	Stored evidence	Auditable AI
Deployment	Docker boundaries	Reproducible runtime

5 ENGINEERING DEEP DIVE

Engineering Metrics

Technical scale supports the claim that PEAXIS is a platform, not a prototype.

16

API modules

Including AI, queue, billing, and tenancy

1

Durable AI queue

ai-tasks with a dedicated worker

5

Canonical work types

Parse, assess, index, and generate

2

Provider options

Gemini or Azure OpenAI per deployment

4

Web apps

Core, Hire, Jobs, and Admin

1

State authority

PostgreSQL and NestJS writes

1

Inference boundary

Stateless FastAPI service

0

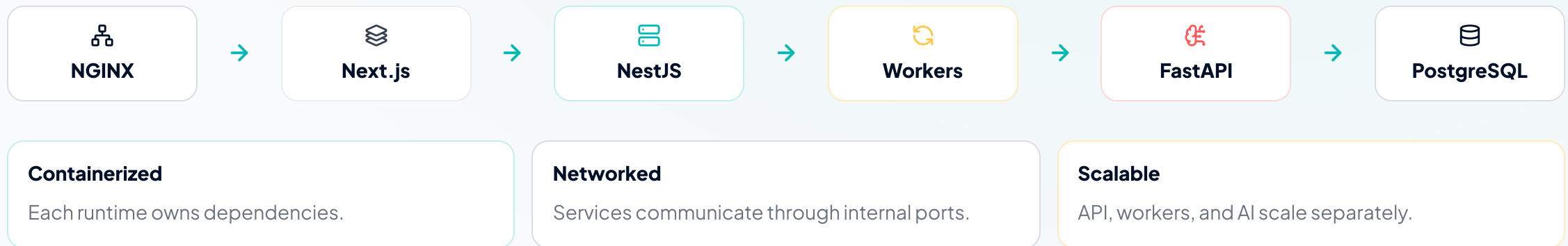
Autonomous rejections

Recruiter review remains mandatory

5 ENGINEERING DEEP DIVE

Deployment Architecture

Container boundaries make the runtime easier to deploy, isolate, and scale.



5 ENGINEERING DEEP DIVE

Future Technical Roadmap

Future work extends the existing architecture instead of replacing it.



RAG

Grounded answers on company and candidate data



AI Agents

Guarded workflow automation



MCP

Standard tool integration



Realtime

Collaborative hiring decisions



Calendar

Interview scheduling automation



Email AI

Candidate communication drafting



Analytics

Hiring funnel and quality metrics



Recommendations

Smarter candidate-job ranking

Modular APIs

Queues

pgvector

Tenant isolation

PEAXIS: Achievement & Future

16

API Modules

1

Durable AI Queue

5

AI Work Types

0

Autonomous Rejections



Key Achievements

- Full-stack multi-tenant SaaS: Core workspace, Hire ATS, and Jobs portal
- Evidence-based assessment: requirements, cited evidence, review, and overrides
- Semantic retrieval: pgvector for public job discovery and evidence retrieval
- Durable processing: AiWorkItem records and the dedicated ai-tasks worker



Technical Lessons

- Schema-first isolation: Multi-tenant guards must be built-in, not retrofitted
- Clear ownership: FastAPI infers; NestJS and PostgreSQL own business state
- Bounded AI: provider selection, credit accounting, and human review control risk



Future Roadmap

- Collaborative workspaces: Real-time decision tracking for hiring teams
- Regional expansion: Multilingual CV parser optimized for the MENA market
- Enterprise connectors: Direct integration with LinkedIn, Greenhouse, and Workday



PEAXIS demonstrates that AI-assisted recruitment can remain reviewable, tenant-safe, and operationally bounded — with people retaining authority over hiring decisions.

Thank you For your time

Contact: BIBANI Mohamed Habib Allah — contact@peaxis.com · Questions welcome
after the session.

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